

Lab 3: Substitutes for Prediction

ECON 470 — Microeconomics of Artificial Intelligence

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Due: TBD at 11:59pm on Canvas. One PDF per group.

Groups: Same groups as Lab 2 (2–3 students, at least one comfortable with calculus).

Estimated time: 3–4 hours.

What you need: Gans, Chapter 4 (Substitutes for Prediction). A calculator or computer for arithmetic. No coding required.

1 Risk Management Without Prediction

A firm operates in a 2×2 environment. There are two possible states of the world, $\theta \in \{1, 2\}$, and two possible actions, $a \in \{1, 2\}$. The firm earns $R = 100$ if its action matches the state and $r = 20$ if it does not. The prior probability that $\theta = 1$ is $p = 0.7$.

The firm can invest in two forms of risk management:

- **Protection** $x \geq 0$: shifts the probability of $\theta = 1$ from p to $p + x$, at cost $C(x) = \frac{\gamma}{2}x^2$ with $\gamma = 600$.
- **Insurance** $\Delta \geq 0$: increases the mismatch payoff from r to $r + \Delta$, at cost $c(\Delta) = \frac{\kappa}{2}\Delta^2$ with $\kappa = 0.01$.

Without any prediction, the firm follows a **rule**: always choose $a = 1$ (the action matching the more likely state).

1.1 Part A: Following a rule with risk management

- (a) Write down the firm's expected payoff as a function of x and Δ when it follows the rule $a = 1$ always. Your expression should include the payoffs from matching and mismatching, minus the costs of protection and insurance.

- (b) Take the first-order condition with respect to x . Show that it gives:

$$R - r - \Delta = \gamma x.$$

Interpret this condition in one sentence: what is the marginal benefit of protection, and what is the marginal cost?

- (c) Take the first-order condition with respect to Δ . Show that it gives:

$$1 - p - x = \kappa \Delta.$$

Interpret this condition in one sentence.

- (d) Solve the system of two equations from (b) and (c) for x_{RULE}^* and Δ_{RULE}^* . You should get $x_{RULE}^* = 0.1$ and $\Delta_{RULE}^* = 20$.
- (e) Compute the firm's **expected payoff under the rule** with optimal risk management. That is, plug x_{RULE}^* and Δ_{RULE}^* back into your expression from (a).
- (f) Verify that insurance and protection are **substitutes** in this model by checking the sign of the cross-partial derivative $\frac{\partial^2 \pi}{\partial x \partial \Delta}$, where π is the expected payoff. Explain in one sentence what this means economically.

2 AI Prediction Arrives — But Is It Good Enough?

Now suppose an AI prediction system becomes available. The system provides a signal $s \in \{1, 2\}$ that correctly identifies the state with probability $e = 0.75$. Using the AI costs $\lambda = 5$ per period.

If the firm were to adopt AI, it would switch from a **rule** to a **decision**: choosing $a = s$ (following the signal). Under a decision, the firm no longer invests in protection ($x_{DEC}^* = 0$, from Proposition 4.1).

2.1 Part B: A signal that beats the prior but not the rule

- (a) Verify that $e > p$. The signal is more accurate than the prior. Based on Chapter 3 alone (ignoring risk management), you might expect this signal to be worth adopting.
- (b) Write down the firm's expected payoff as a function of Δ when it follows the signal, with $x = 0$, and pays the AI cost λ :

$$eR + (1 - e)(r + \Delta) - c(\Delta) - \lambda.$$

- (c) Take the first-order condition with respect to Δ and solve for Δ_{DEC}^* .
- (d) Compute the firm's **expected payoff under the decision** with optimal insurance and AI cost.
- (e) Compare your answer from (d) to the rule payoff from Problem 1(e). Should the firm adopt AI? Why or why not?

- (f) Check the condition $e > p + x_{RULE}^*$. Does it hold? Explain in 2–3 sentences why this condition — not just $e > p$ — is the relevant threshold for adoption when the firm has invested in protection.
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3 Better AI: Prediction Adopted

The firm now has access to a better AI system with accuracy $e = 0.85$ at the same cost $\lambda = 5$.

3.1 Part C: A signal that clears the bar

- Verify that $e > p + x_{RULE}^*$ now holds.
 - Solve for Δ_{DEC}^* with $e = 0.85$.
 - Compute the firm’s **expected payoff under the decision** with this better signal.
 - Should the firm adopt AI now? By how much does the expected payoff change relative to the rule?
 - Compare Δ_{DEC}^* under the better signal to $\Delta_{RULE}^* = 20$. Did insurance increase or decrease? Is this consistent with Proposition 4.1? Explain in 1–2 sentences.
 - Verify that the necessary condition from Corollary 4.1 ($e > p$) holds. Explain in one sentence why this condition is necessary but, as Problem 2 showed, not sufficient.
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4 Can Insurance Ever Rise When AI Is Adopted?

Proposition 4.1 states that insurance is reduced if e is sufficiently high, but may increase if e is sufficiently low. In Problems 2 and 3, you computed Δ_{DEC}^* for two values of e .

4.1 Part D: A conceptual question

- In Problem 2, you found $\Delta_{DEC}^* > \Delta_{RULE}^*$: the decision-maker would want *more* insurance than under the rule. In Problem 3, you found $\Delta_{DEC}^* < \Delta_{RULE}^*$: the decision-maker wants less. Confirm these two inequalities using your earlier answers.
- Notice the pattern: when e is low enough that insurance would rise ($e < p + x_{RULE}^*$), the firm does not adopt AI. When e is high enough that the firm adopts AI ($e > p + x_{RULE}^*$), insurance falls. Explain in 2–3 sentences why this is not a coincidence — why does the adoption threshold $e = p + x_{RULE}^*$ also separate the “insurance rises” and “insurance falls” regions?

- (c) Proposition 4.1 says insurance *can* rise under a decision. But your analysis shows that a firm choosing optimally between a rule and a decision will never actually experience rising insurance, because it only adopts AI when e is high enough for insurance to fall. In 2–3 sentences, explain what Proposition 4.1 is really telling us. What kind of comparison is it making, if not a comparison of outcomes the firm would actually choose?
- (d) Give a real-world example where an organization might be *forced* to switch from a rule to a decision (perhaps by regulation, competition, or a mandate) at a signal quality where $e < p + x_{RULE}^*$. In that case, would you expect the organization's insurance spending to rise or fall? Why?